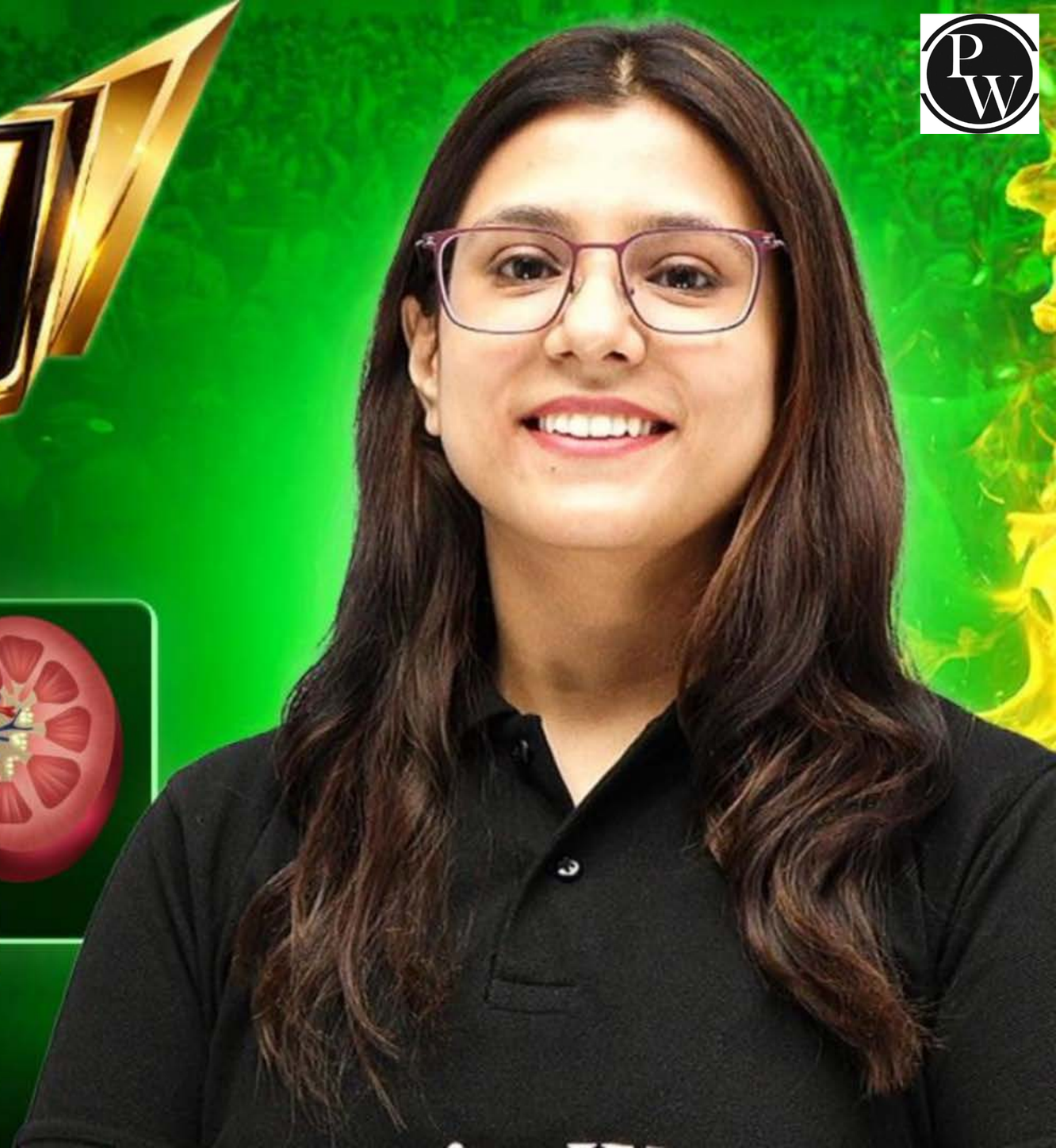
VICTORY 2027

The Excretory System



LECTURE-1 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Blood Group , Erythroblastosis foetalis
2. Excretion


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

RATTON KO KARO **BYE BYE !!**

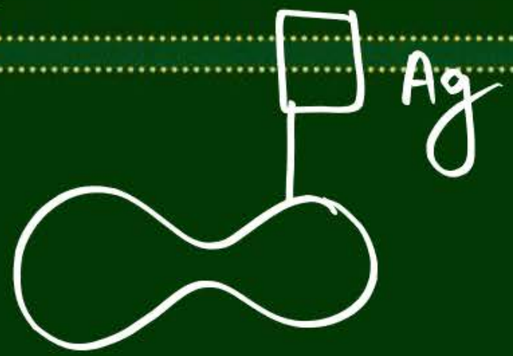
EASY CONCEPTS SE KARO **HELLO HI !!**

With – *Uppu Ma'am*



QUESTIONS

The substances present on the surface of RBCs that determine blood group are called:



A Antibodies

B Antigens

C Platelets

D Plasma proteins

Am I wearing glasses?

A) Yes um ✓

B) No um

QUESTIONS

A person with blood group A has which antigen on RBCs?

A. Antigen A ✓

B. Antigen B

C. Both (A) and (B)

D. No antigen

QUESTIONS

A person with blood group B has:

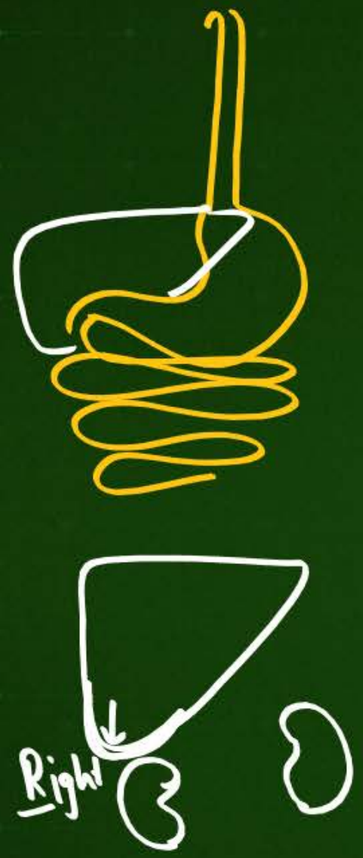
Ag Ab

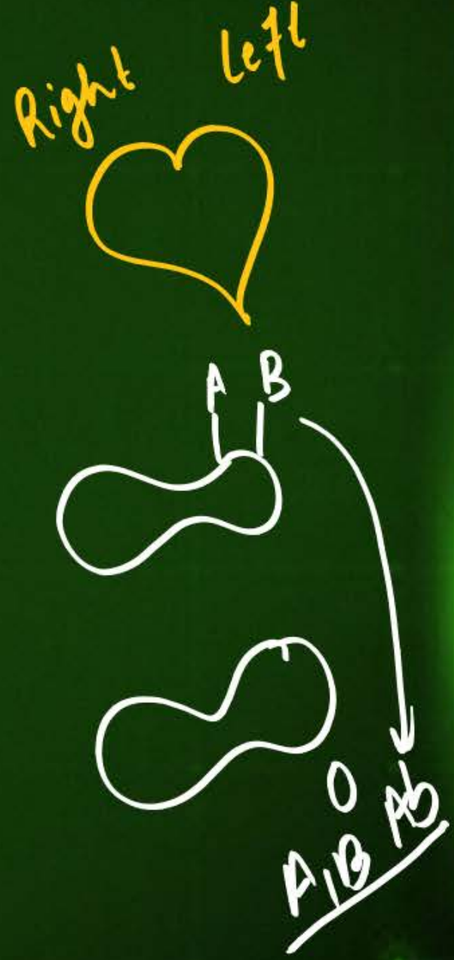
A. Antigen A and anti-B antibody

B. Antigen B and anti-A antibody ✓

C. Both antigens and no antibodies

D. No antigen and no antibodies


Right


Right
Left
A
B
O
A B AB

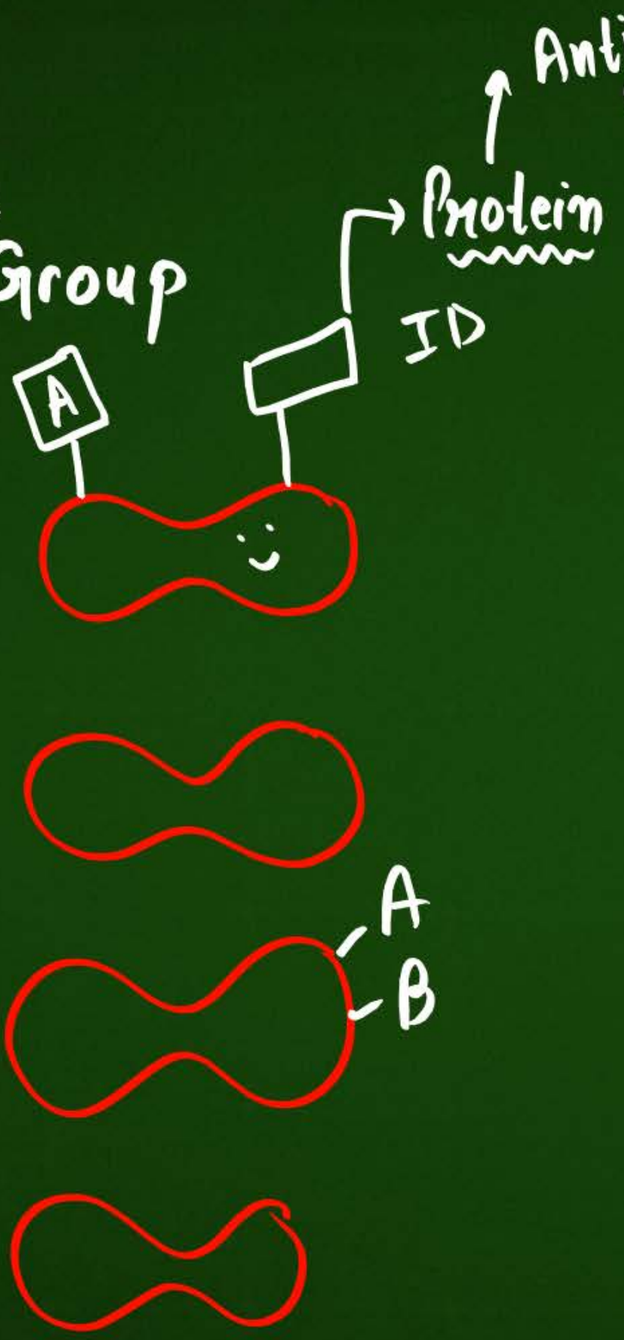
BLOOD TRANSFUSION AND BLOOD GROUPS (ABO AND RH SYSTEMS)

- The concept of blood grouping was discovered by Karl Landsteiner. RBCs of human beings have specific proteins on their surface.
- These proteins are called antigens.
- In humans, there are two types of antigens.

1. ANTIGEN A
2. ANTIGEN B

- The plasma of the blood contains complementary proteins, called antibodies, with respect to the antigen that is present on the surface of RBCs.

Tumhara Blood Group



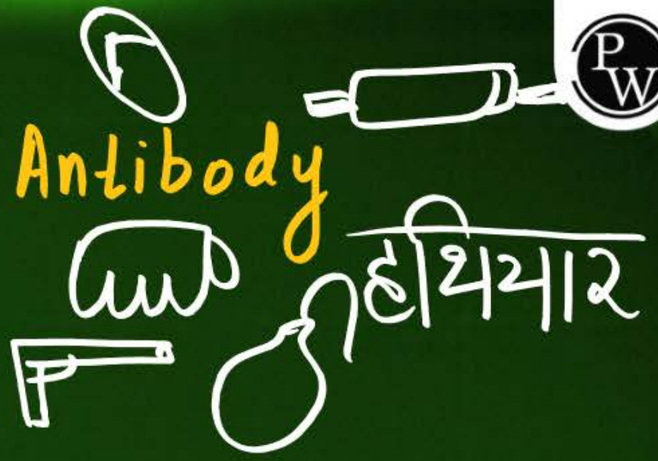


A - Antigen

B Antigen

A, B Antigen

No Antigen

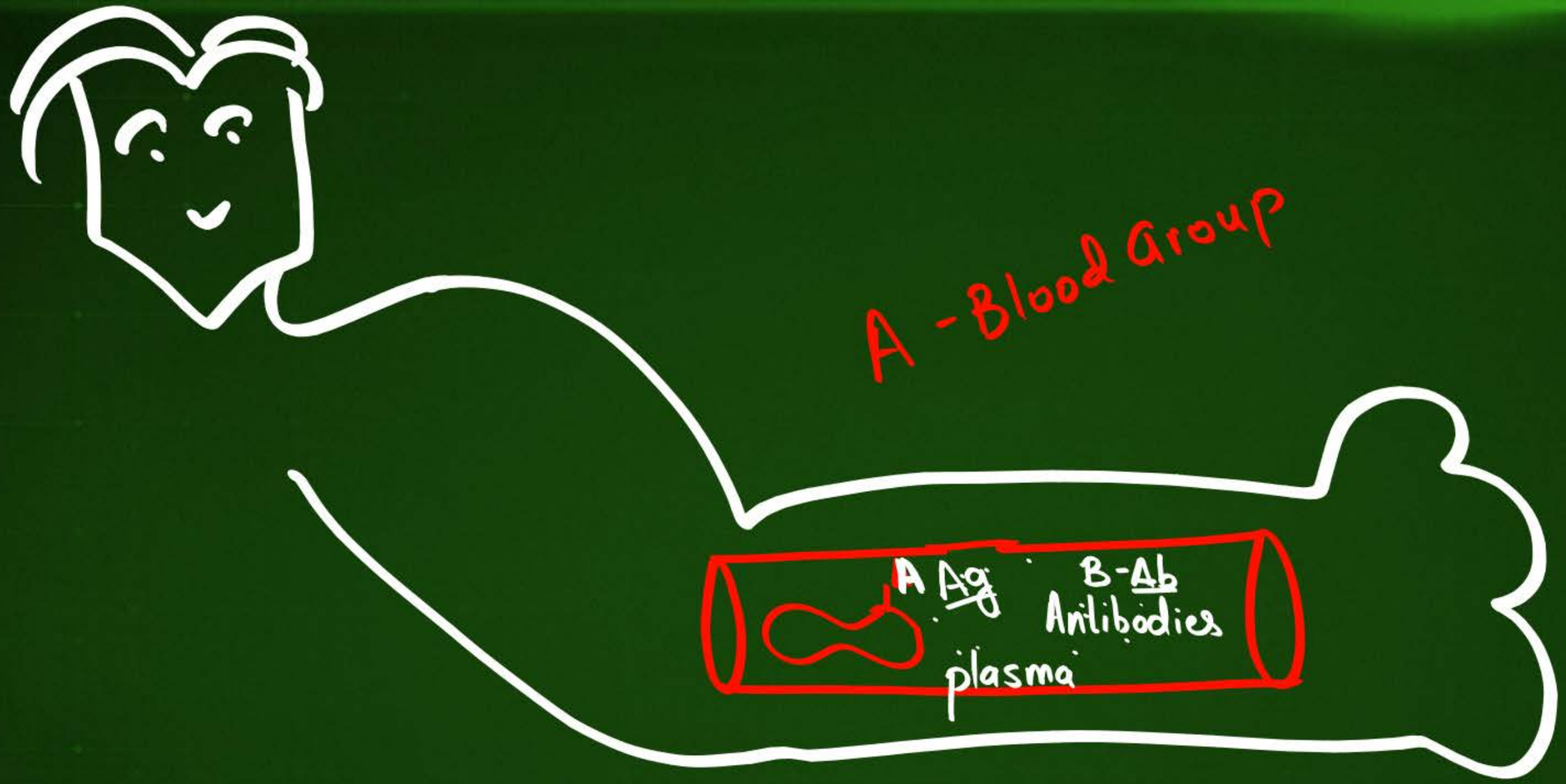


B- Antibody

A- Antibody

No Antibody

A, B Antibody


A - Blood Group
A Ag
B-Ab
Antibodies
plasma

SEVERAL SYSTEM OF BLOOD GROUPING

Blood group	Antigens on RBC	Antibodies in plasma
A	Antigen A	Antibody B
B	Antigen B	Antibody A
AB	Antigen A and B	No antibody
O	No antigen	Both antibody A and antibody B

SUMMARY OF ABO BLOOD GROUP MATCHING (COMPATIBILITY)

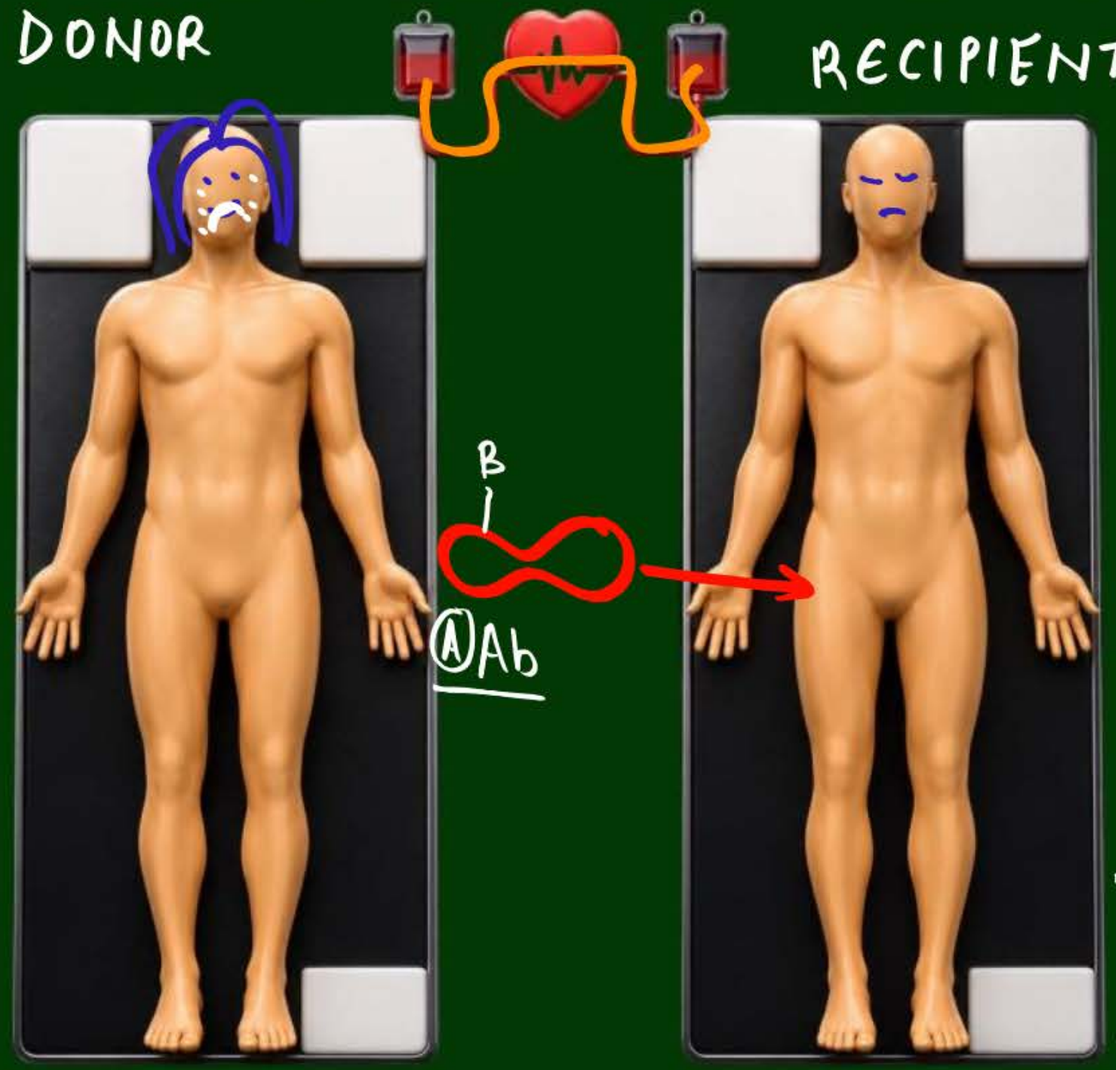
Blood Group of Donor	Blood Group of Recipient			
	A	B	AB Universal Recipient	O
A	✓	✗	✓	✗
B	✗	✓	✓	✗
AB	✗	✗	✓	✗
O Universal Donor	✓	✓	✓	✓

BLOOD TRANSFUSION

WRONG X

DONOR

RECIPIENT



5 Hr RIP

Uppu mummy

B- Ag

A- Ab

Blood Group

A- Antigen (Ag)

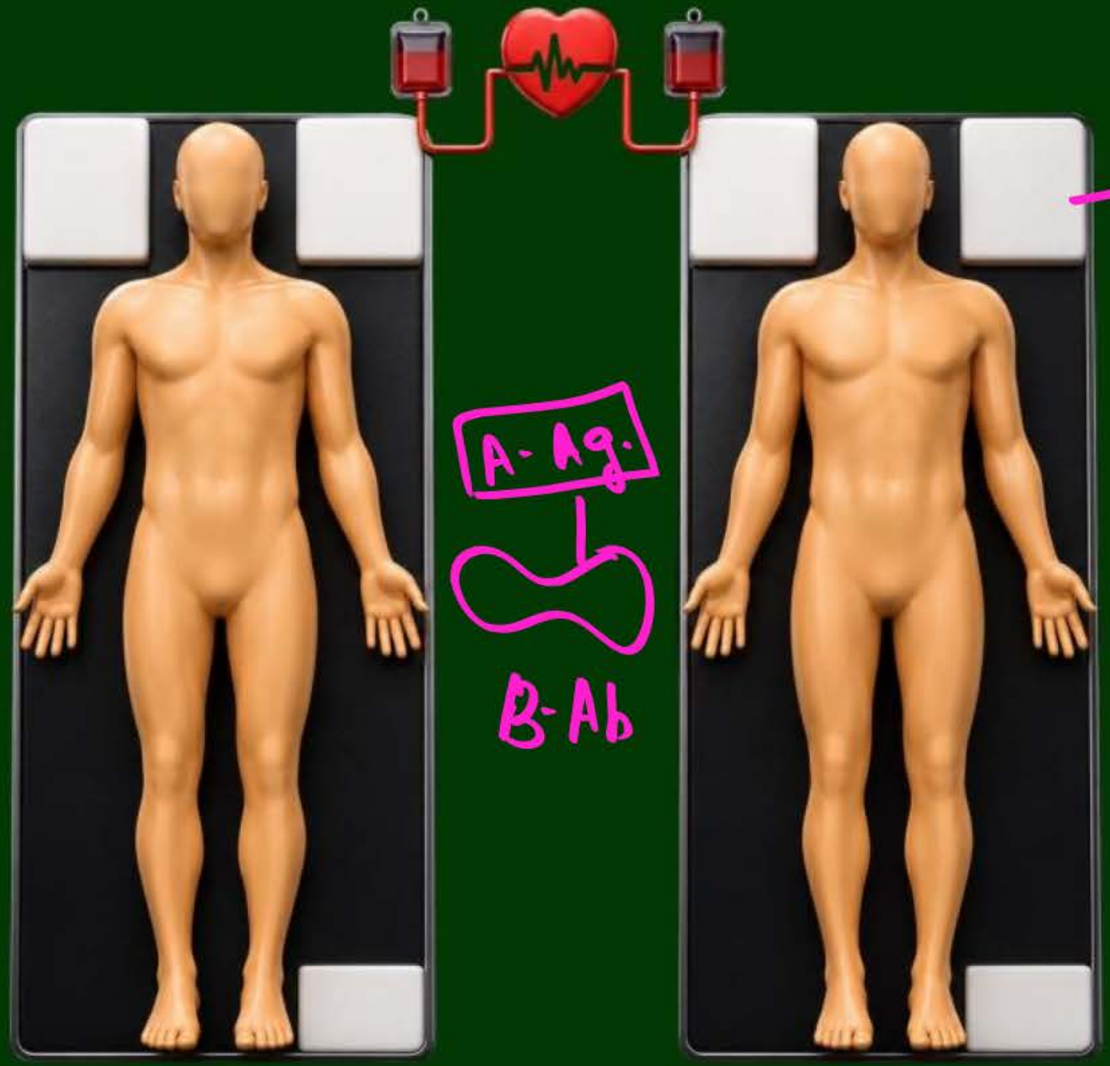
B- Antibody (Ab)

Same Ag-Ab ENEMIES)

'Ab' from Donor will be diluted in the Recipient's blood (No Danger)

UMC

A - Ag.
B - Ab



→ A

A - Ag.
B - Ab

✓

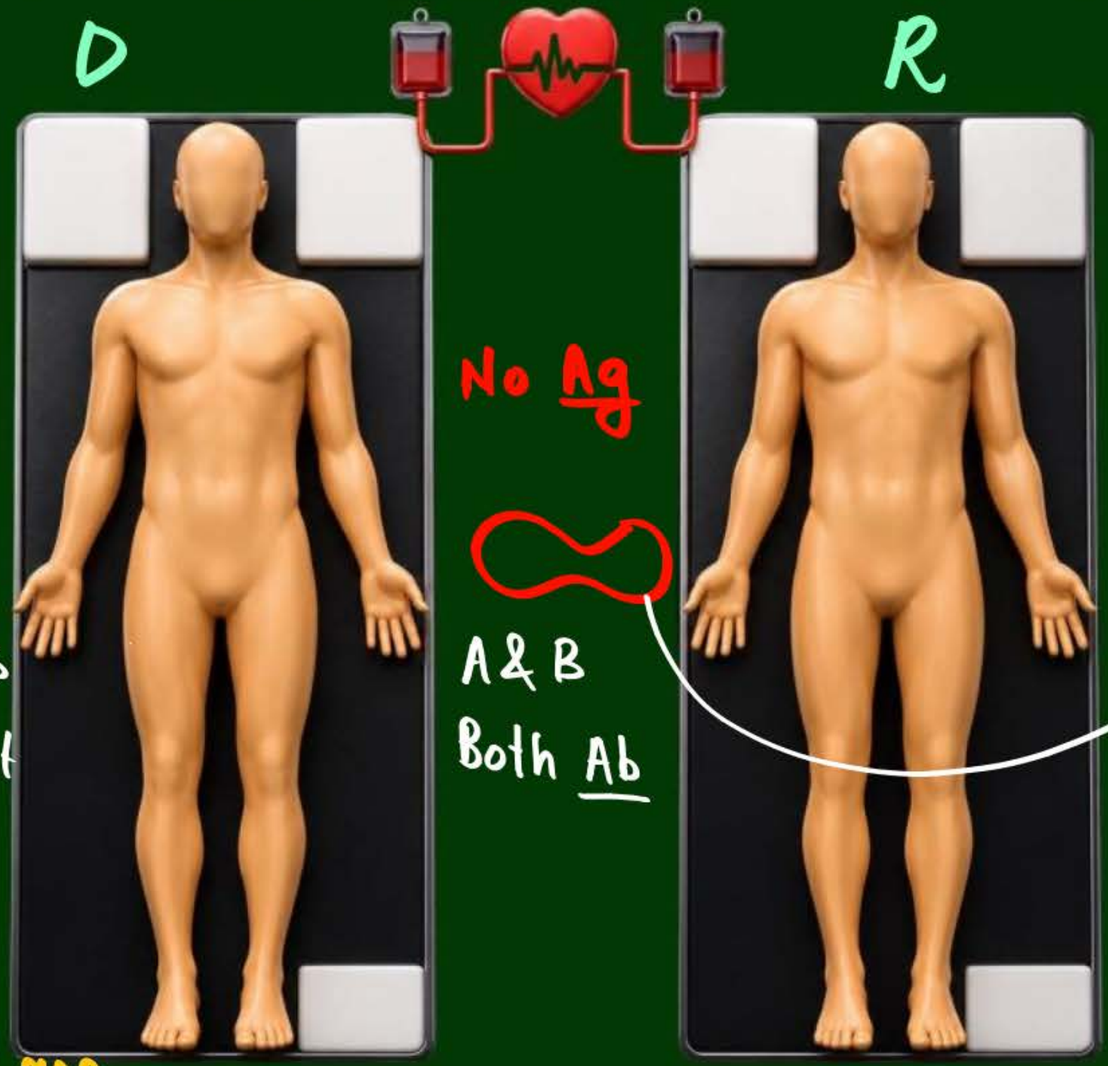
B-Ab & A-Ag.

Successful Blood Transfusion

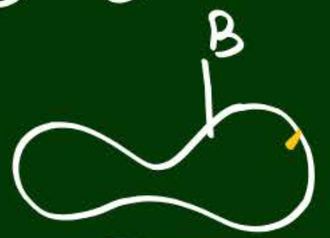
No RBC coagulation

O Blood Group

- Universal Donor
- Because, the RBC has no Antigen, the Recipient's Antibody will not react with any Antigen.
- It can accept blood only from O Blood grp.



B - Blood Group



A - Blood group





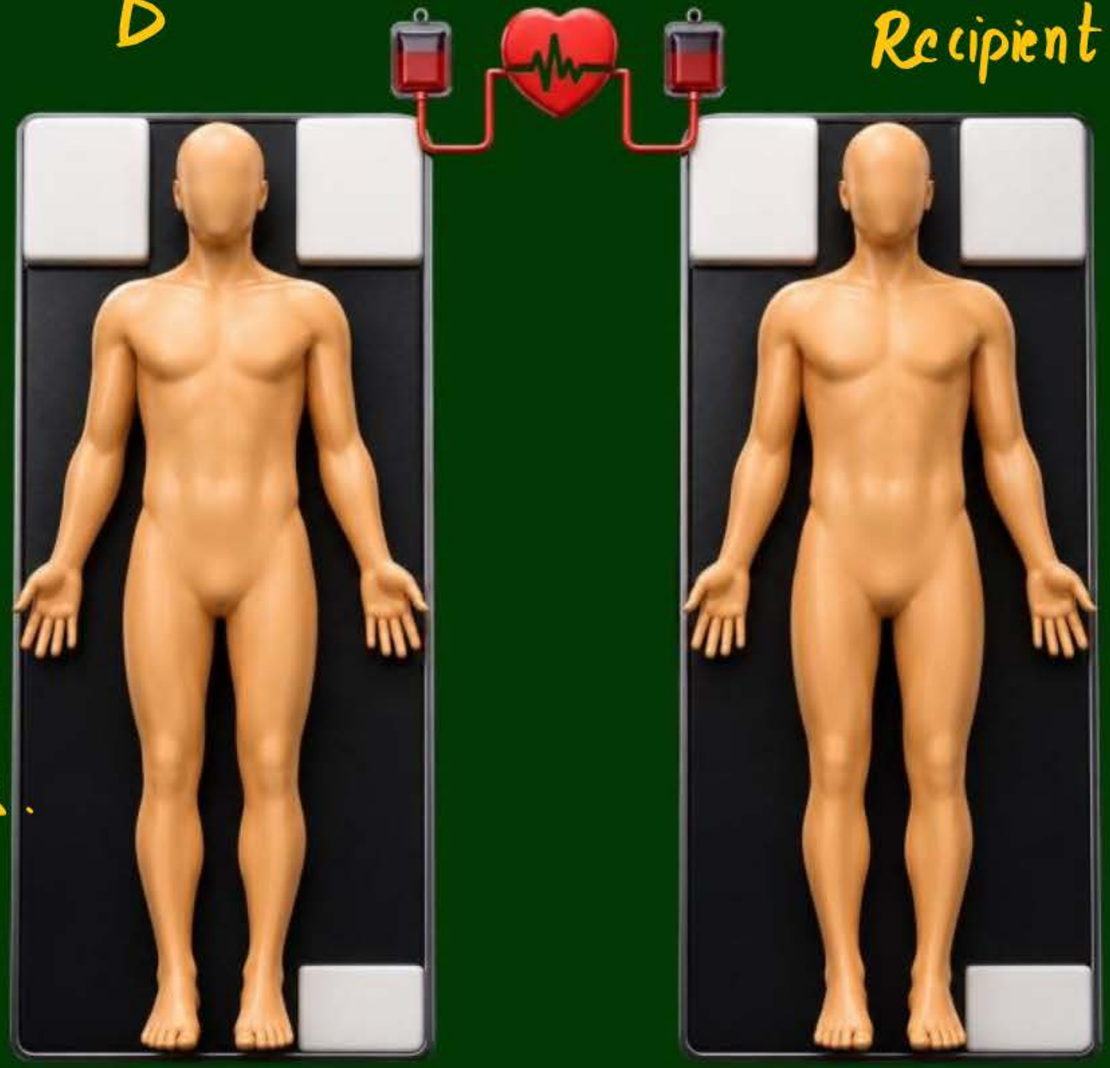
AB Blood Group
No Ab

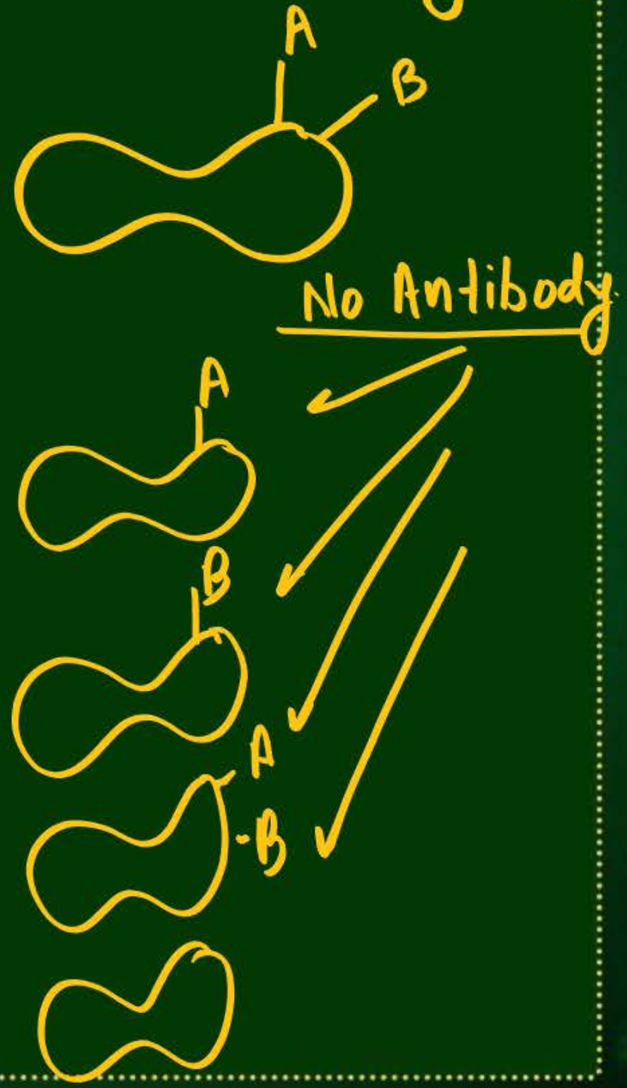


O Blood Group
A, B Ab

AB Universal Recipient

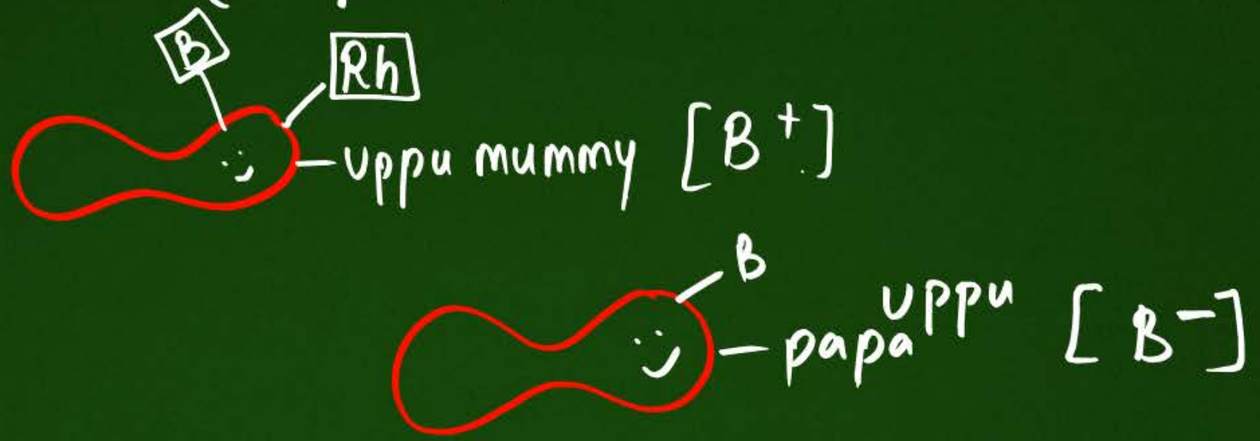
- They don't have Antibodies to fight Antigen, so they can receive blood from all Blood Groups.



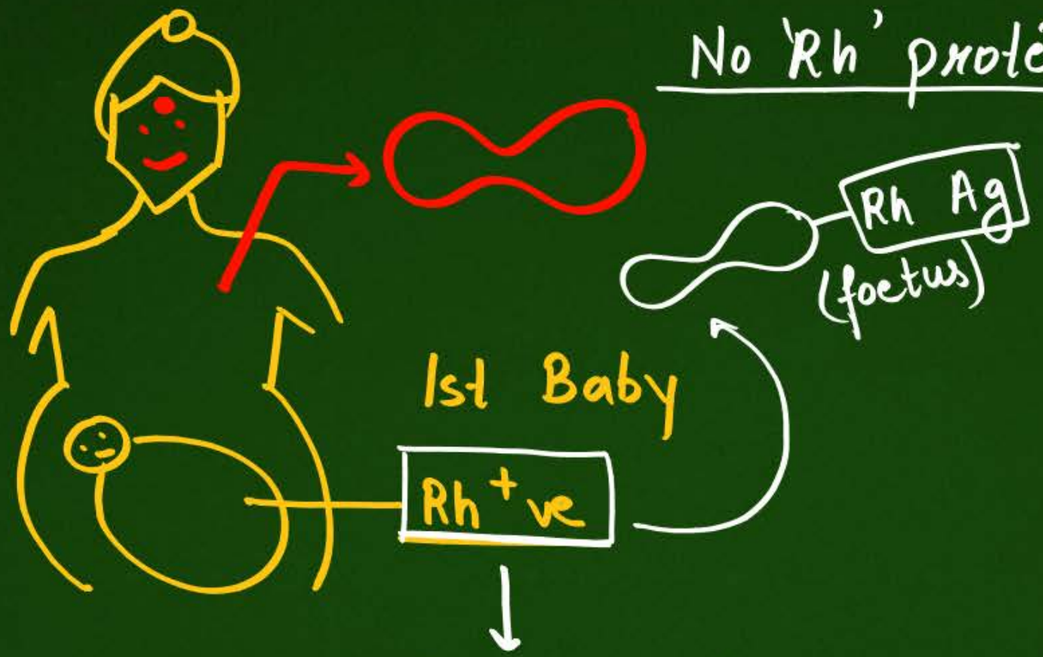


Rhesus factor
(Rh factor)

- Rh is also an Antigen
- In 85% population 'Rh' factor is present & in rest 15% population 'Rh' factor is absent.



Mother [-ve] Father [+ve]

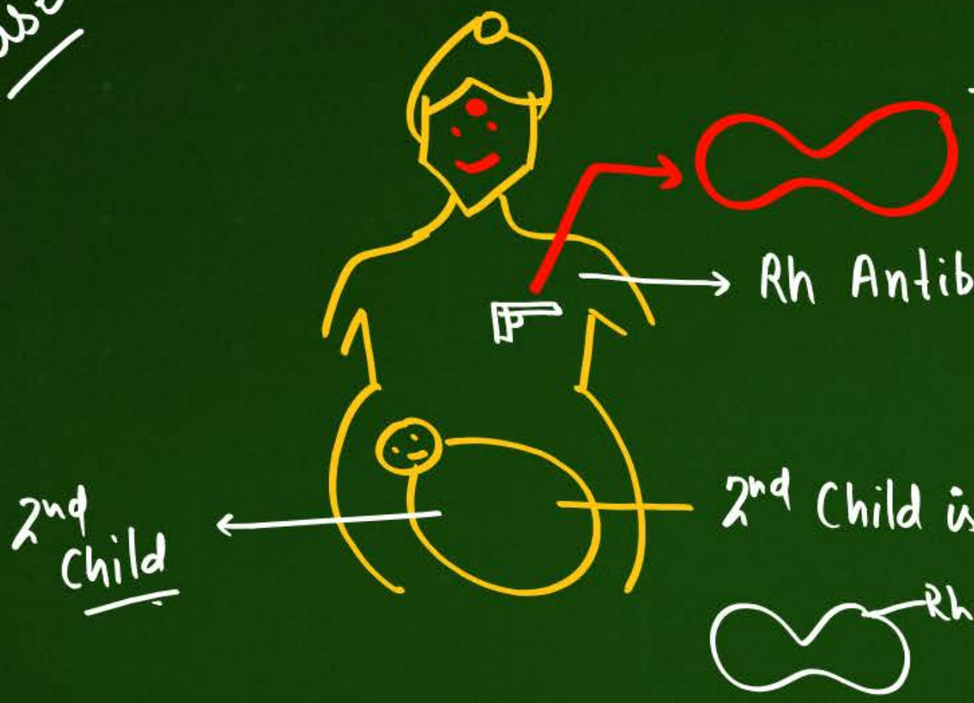


At the time of delivery, some blood from child can mix with mother's blood → Mother's blood will start forming 'Rh' Ab because the foetus blood was Rh+ve.

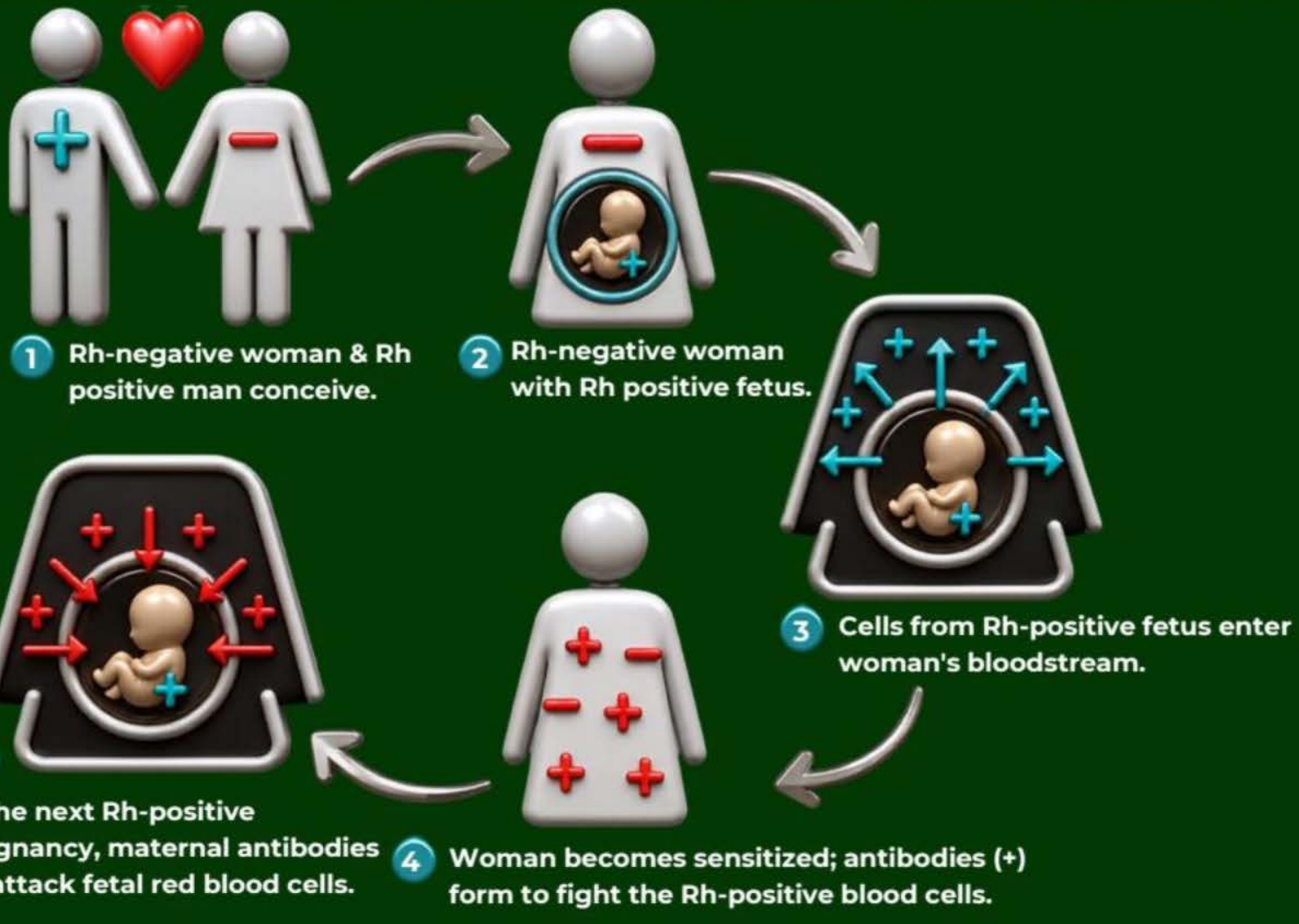
IInd case

Mother [-ve] Father [+ve]

No 'Rh' protein [Mother]



'Rh' Ab from mother can cross through Placenta and will attack RBC of foetus. → This will result in coagulation of RBC of foetus. Hence the condition ERYTHROBLASTOSIS FOETALIS


1 Rh-negative woman & Rh positive man conceive.
2 Rh-negative woman with Rh positive fetus.
3 Cells from Rh-positive fetus enter woman's bloodstream.
4 Woman becomes sensitized; antibodies (+) form to fight the Rh-positive blood cells.
5 In the next Rh-positive pregnancy, maternal antibodies (+) attack fetal red blood cells.

Excretion

The process of removal of waste from the body like CO_2, Urea, Uric acid, Creatinine, extra salts and water.

ammonia

UREA is the main excretory waste, it is produced in the LIVER.

"WHO RUINED THE BLOOD?"

- Urea
- Glucose
- Protein
- Na^+
- Water



Thank You